

MAXIMILIANO RODRIGUEZ

ROBOTICS ENGINEER • EMBEDDED SYSTEMS • AI SOLUTIONS ARCHITECT

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PROFESSIONAL SUMMARY

High-performance robotics engineer and software developer with 4+ years of competitive international experience. **Ranked #19 Worldwide (WRO 2023)** and selected as **2026 World Finalist** representing the Venezuelan National Delegation. Demonstrated technical expertise in embedded C++/C# architecture, autonomous multi-sensor platforms (Raspberry Pi, Arduino), and real-time signal processing. Founder of **Paper Vortex Studio**, engineering commercial AI automations, custom business web infrastructure, and interactive software systems.

HONORS & COMPETITIVE AWARDS

World Robot Olympiad (WRO) International Finalist (Panama City, Panama) Ranked #19 Worldwide and #4 in the Americas representing Venezuela; competed among elite global robotics delegations.	Nov 2023
World Robot Olympiad (WRO) International Finalist (San Juan, Puerto Rico) Selected to represent Venezuela at the World Finals with the ORBIT autonomous AI social robotics system.	2026
WRO Venezuela National Championship (Caracas, Venezuela) Secured a Top 4 National Ranking in the high-stakes senior robotics division.	2024
Kurios Robotics Championship (Puerto La Cruz, Venezuela) Awarded 1st Place Overall (2022, 2023) and 2nd Place Overall (2024, 2025) across 4 consecutive competition seasons.	2022 – 2025

ENGINEERING SYSTEMS & COMMERCIAL VENTURES

ORBIT – Autonomous Social Robot Golden Codex (WRO Team) - Architected an autonomous social robot on Raspberry Pi 5 equipped with a custom 4-motor chassis and animated facial telemetry. - Engineered a real-time acoustic trilateration localization system using 3 directional microphones to triangulate human speakers. - Integrated conversational AI to guide public discourse toward culture and STEM; modeled as an interactive kiosk for museums.	Oct 2025 – Present
Founder & Principal Solutions Developer Paper Vortex Studio - Founded a commercial technology studio delivering enterprise AI integrations, custom web architecture, and interactive software. - Engineered 24/7 AI conversational automation for real estate and business sales, integrating WhatsApp APIs with live property databases for instant inventory inquiries, intelligent lead qualification, and CRM logging (papervortexstudio.com). - Designs, deploys, and manages high-performance web infrastructure and automated digital workflows for emerging entrepreneurs. - Directs an experimental game mechanics lab; designed procedural state machines and physics in Unity (C#) for 'The Other Side'.	Aug 2024 – Present
Automated Maritime Tide Monitoring System Tomorrow's Innovators (WRO Team) - Prototyped a real-time coastal flood hazard alert system on Arduino UNO targeting maritime safety and coral reef conservation. - Programmed analog signal filters in C++ across multi-depth submersible sensor terminals to trigger instant color-coded LED warnings. - Built and demonstrated a closed-loop fluid simulation apparatus driven by automated water pumps to validate variable tidal spikes.	Mar 2023 – Dec 2023
Autonomous Seed-to-Microgreen Agricultural System Tomorrow's Innovators (WRO Team) - Engineered a fully autonomous indoor agricultural robot executing an 8-day seed-to-harvest growth cycle aligned with UN SDG-12. - Integrated Arduino UNO, Raspberry Pi 4, stepper motors, servos, and programmed timed LED grow lighting schedules in C++.	Jan 2024 – Dec 2024

TECHNICAL PROFICIENCIES

Programming & Software:	C++, C#, Python (Certified), HTML5/CSS/JavaScript, Bash/Shell Scripting, Git/GitHub
AI & Enterprise Integration:	LLM API Integrations, WhatsApp Business Automation, Conversational Agents, CRM & Database Sync
Embedded Hardware & IoT:	Raspberry Pi (4 & 5), Arduino (UNO), Steppers, Servos, Analog Sensors, Circuit Prototyping
Core Specialties:	Autonomous Robotics, Acoustic Trilateration, Real-Time Signal Processing, Full-Stack Web Infrastructure

EDUCATION, TRAINING & ATHLETICS

Colegio Ítalo Venezolano 'Angelo De Marta' — High School Diploma Academic Standing: 10th Grade (Sophomore). Consistent top-tier academic marks in Mathematics, Physics, and Chemistry. Supplemental Technical Training: Kurios Academy (Robotics & Strategy, 2022–Pres.); Tech Academy (Advanced Prototyping, 2025–Pres.); ADAKADEMY (Python Certification, 2022). Athletics & Discipline: Angelo Escudero Taekwondo Academy (2018–Pres.) — Red Belt with 1 Black Stripe; 8 consecutive years of competitive training. Languages: Spanish (Native), English (C1 – Advanced Professional Proficiency), French (A2 – Elementary).	Expected June 2027
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